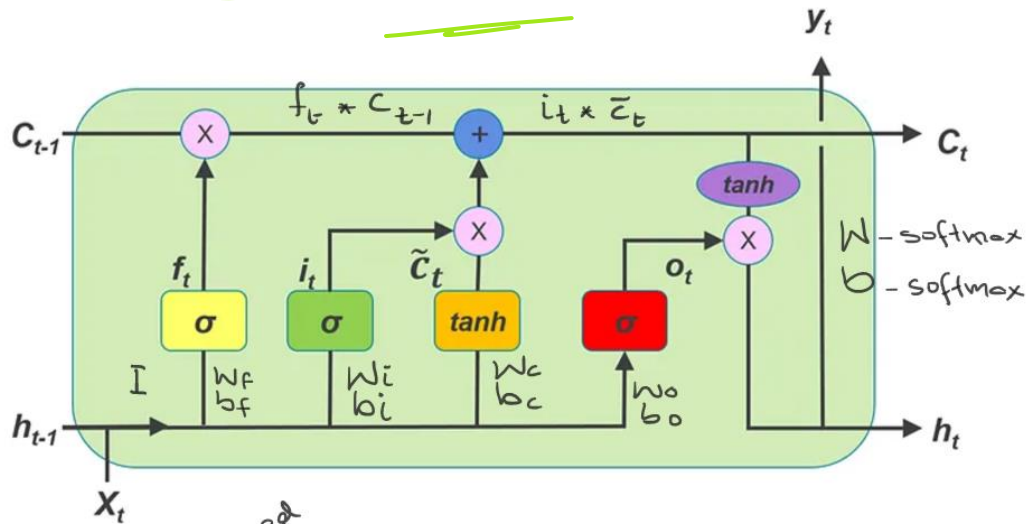


Solution



• Reduced Equations For LSTM

$$I = [h_{t-1}, x_t]$$

$$f_t = \text{sigmoid}(W_f * I + B_f)$$

$$i_t = \text{sigmoid}(W_i * I + B_i)$$

$$o_t = \text{sigmoid}(W_o * I + B_o)$$

$$\tilde{c}_t = \tanh(W_c * I + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{c}_t$$

$$h_t = o_t * C_t$$

Q: We've thoroughly practiced employing LSTM (Long Short-Term Memory) in our previous assignment to forecast forthcoming work tasks. The current objective involves computing values for the below given tasks.

1. Compute embedding from the given target weight matrix based on One Hot vector: [1 0 0 0].
2. Define Stacked Input.
3. Compute value for forget gate from the data given below.
4. Compute C_t & h_t value from all supporting values given below.
5. Write Equations for finding C_t & h_t .

Target Weight Matrix:

4	1	3	4
2	3	3	4
4	1	1	0
2	0	2	4

Weight Matrix for Forget Gate:

6	2	4	6	6	4	4	5
5	5	1	1	0	5	6	4
2	4	2	0	1	5	5	5
6	4	2	3	1	6	3	6

Bias for Forget Gate:

0
0
2
0

Name: _____

Quiz

Roll#: _____

Forget Gate

1
1
1
1

Input Gate

1
1
1
0.99

Output Gate

1
1
0.99
1

 h_{t-1}

0
0
0
0

 c_{t-1}

0
0
0
0

 $\tilde{c}_t$

1
0
0
0

Solution: (Show Steps)

Insert matrix Restore matrix

	A ₁	A ₂	A ₃	A ₄
1	4	1	3	4
2	2	3	3	4
3	4	1	1	0
4	2	0	2	4

Complex numbers (more)

Fractional ▾

	B ₁
1	1
2	0
3	0
4	0

	C ₁
1	4
2	2
3	4
4	2

Insert matrix Restore matrix

	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	A ₇	A ₈
1	6	2	4	6	6	4	4	5
2	5	5	1	1	0	5	6	4
3	2	4	2	0	1	5	5	5
4	6	4	2	3	1	6	3	6

Clear Fill empty cells with zero

on: 8 X 1

Fractional ▾

	B ₁
1	0
2	0
3	0
4	0
5	4
6	2
7	4
8	2

	C ₁
1	58
2	42
3	44
4	40

$$c_t = \begin{bmatrix} f_t \\ 1 \\ 1 \\ 1 \end{bmatrix} \odot \begin{bmatrix} c_{t-1} \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} i_t \\ 1 \\ 1 \\ 0.99 \end{bmatrix} \odot \begin{bmatrix} \tilde{c}_t \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} c_t \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$h_t = \begin{bmatrix} 1 \\ 1 \\ 0.99 \\ 1 \end{bmatrix} \odot \begin{bmatrix} c_t \\ 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Name: _____

Quiz

Roll#: _____

Q: We've thoroughly practiced employing LSTM (Long Short-Term Memory) in our previous assignment to forecast forthcoming work tasks. The current objective involves computing values for the below given tasks.

1. Compute embedding from the given target weight matrix based on One Hot vector: [0 1 0 0]
2. Compute value for forget gate from the data given below.
3. Compute C_t & h_t value from all supporting values given below.
4. Write Equations for finding C_t & h_t .

Target Weight Matrix:

4	1	3	4
2	3	3	4
4	1	1	0
2	0	2	4

Weight Matrix for Input Gate:

6	4	3	1	5	6	2	0
0	0	6	6	1	3	3	5
6	4	2	3	4	5	1	2
6	6	2	6	1	4	0	0

Bias for Input Gate:

4
4
2
1

Forget Gate

1
1
1
1

Input Gate

1
1
1
1

Output Gate

1
1
1
1

 h_{t-1}

0.76
0.76
0.76
0.76

 c_{t-1}

1
1
1
0.99

 $\tilde{c}_t$

0
1
0
0

Solution: (Show Steps)

$$f_t = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0.99 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0.99 \end{bmatrix} \quad \text{(Circled } C_t \text{)}$$

$$h_t = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0.99 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0.99 \end{bmatrix} \quad \text{(Circled } h_t \text{)}$$

Name: _____

Quiz

Roll#: _____

Q: We've thoroughly practiced employing LSTM (Long Short-Term Memory) in our previous assignment to forecast forthcoming work tasks. The current objective involves computing values for the below given tasks.

1. Compute embedding from the given target weight matrix based on One Hot vector: [0 0 1 0]
2. Compute value for forget gate from the data given below.
3. Compute C_t & h_t value from all supporting values given below.
4. Write Equations for finding C_t & h_t .

Target Weight Matrix:

4	1	3	4
2	3	3	4
4	1	1	0
2	0	2	4

Weight Matrix for Output Gate:

1	3	0	2	2	4	2	3
0	2	0	2	0	3	6	0
0	4	0	2	0	0	1	3
3	4	1	0	5	1	1	0

Bias for Output Gate:

5
2
3
8

Forget Gate

1
1
1
1

Input Gate

1
1
1
1

Output Gate

1
1
1
1

 h_{t-1}

0.964
0.964
0.964
0.964

 c_{t-1}

2
2
2
1.99

 $\tilde{c}_t$

0
0
1
0

Solution: (Show Steps)

Insert matrix

Restore matrix

A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8
1	3	0	2	2	4	2	3
0	2	0	2	0	3	6	0
0	4	0	2	0	0	1	3
3	4	1	0	5	1	1	0

Clear

Fill empty cells with zero

B_1
0.964
0.964
0.964
0.964
3
3
1
2

C_1
31.784
18.856
12.784
26.712

$$C_t = f_t \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 1.99 \end{bmatrix} + i_t \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 1.99 \end{bmatrix} \quad (C_t)$$

$$h_t = o_t \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \\ 1.99 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 1.99 \end{bmatrix} \quad (h_t)$$